Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:

  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using pandas.
- **Initial Exploration:** Used df.info() to check structure and `.describe()` for summary statistics.

```
 12  Shipping Type          3900 non-null   str
 13  Discount Applied       3900 non-null   str
 14  Promo Code Used        3900 non-null   str
 15  Previous Purchases     3900 non-null   int64
 16  Payment Method         3900 non-null   str
 17  Frequency of Purchases 3900 non-null   str
dtypes: float64(1), int64(4), str(13)
memory usage: 548.6 KB
None
       Customer ID         Age  Purchase Amount (USD)  Review Rating  Previous Purchases
count  3900.000000  3900.000000            3900.000000    3863.000000         3900.000000
mean   1950.500000    44.068462              59.764359       3.750065           25.351538
std    1125.977353    15.207589              23.685392       0.716983           14.447125
min       1.000000    18.000000              20.000000       2.500000            1.000000
25%     975.750000    31.000000              39.000000       3.100000           13.000000
50%    1950.500000    44.000000              60.000000       3.800000           25.000000
```

```
Age                         0
Gender                      0
Item Purchased              0
Category                    0
Purchase Amount (USD)       0
Location                    0
Size                        0
Color                       0
Season                      0
Review Rating               0
Subscription Status         0
Shipping Type               0
Discount Applied            0
Promo Code Used             0
Previous Purchases          0
Payment Method              0
Frequency of Purchases      0
```

- **Missing Data Handling:** Checkedfor null values and imputed missing values in the `Review Rating` column using themedian rating of each product category.

- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.

- **Feature Engineering:**

  - Created **age_group** column by binning customer ages.

  - Created **purchase_frequency_days** column from purchase data.

- **Data Consistency Check:** `Verified if discount_applied and promo_code_used` were redundant; dropped promo_code_used.

- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

```
ysql> select gender,SUM(purchase_amount) as revenue from customer group by gender;
--------+---------+
 gender | revenue |
--------+---------+
 Male   |  157890 |
 Female |   75191 |
--------+---------+
 rows in set (0.02 sec)
```

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

```
mysql> select customer_id , purchase_amount from customer where(discount_applied='yes') and purchase_amount>=(select AVG(purchase_amou
nt) from customer);
+-------------+-----------------+
| customer_id | purchase_amount |
+-------------+-----------------+
|           2 |              64 |
|           3 |              73 |
|           4 |              90 |
|           7 |              85 |
|           9 |              97 |
|          12 |              68 |
|          13 |              72 |
|          16 |              81 |
|          20 |              90 |
|          22 |              62 |
|          24 |              88 |
|          29 |              94 |
|          32 |              79 |
|          33 |              67 |
|          35 |              91 |
|          37 |              69 |
|          40 |              60 |
|          41 |              76 |
|          43 |             100 |
|          44 |              69 |
|          55 |              94 |
|          57 |              73 |
|          58 |              64 |
```

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

```
purchased,ROUND(AVG(review_rating),2) as "Average Review Rating" from customer group by item_purc
c limit 5;
-----------------------+
 Average Review Rating |
-----------------------+
                  3.86 |
                  3.84 |
                  3.82 |
                   3.8 |
                  3.78 |
-----------------------+
 sec)
```

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

```
mysql> select shipping_type, AVG(purchase_amount) from customer where shipping_type in ('Standard
press') group by shipping_type;
+---------------+----------------------+
| shipping_type | AVG(purchase_amount) |
+---------------+----------------------+
| Express       |              60.4752 |
| Standard      |              58.4602 |
+---------------+----------------------+
2 rows in set (0.27 sec)
```

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

```
mysql> select subscription_status , COUNT(customer_id) as total_customers, ROUND(AVG(purchase_amount),
2) as avg_spend, ROUND(AVG(purchase_amount),2) as total_revenue from customer group by subscription_st
atus order by total_revenue,avg_spend desc;
+---------------------+-----------------+-----------+---------------+
| subscription_status | total_customers | avg_spend | total_revenue |
+---------------------+-----------------+-----------+---------------+
| Yes                 |            1053 |     59.49 |         59.49 |
| No                  |            2847 |     59.87 |         59.87 |
+---------------------+-----------------+-----------+---------------+
2 rows in set (0.15 sec)
```

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

item_purchased	discount_rate
Socks	32.70
Blouse	33.92
Sandals	36.88
Skirt	38.61
Handbag	39.87

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

```
ysql> with customer_type as (select customer_id,previous_purchases, CASE WHEN previous_purchases = 1 THEN 'New' WHEN previou
_purchases BETWEEN 2 AND 10 THEN 'Returning' ELSE 'Loyal' END AS customer_segment from customer) select customer_segment, co
nt(*) as "Number of Customer" from customer_type group by customer_segment;
-----------------+--------------------+
 customer_segment | Number of Customer |
-----------------+--------------------+
 Loyal            |               3116 |
 Returning        |                701 |
 New              |                 83 |
-----------------+--------------------+
 rows in set (0.02 sec)
```

8. **Top3Products per Category** – Listed the most purchased products within each category.

```
by category order by count(customer_id) DESC) as item_rank from customer group by category,item_purchased) select item_rank,c
ategory,item_purchased,total_orders from item_counts where item_rank<=3;
+-----------+-------------+----------------+--------------+
| item_rank | category    | item_purchased | total_orders |
+-----------+-------------+----------------+--------------+
|         1 | Accessories | Jewelry        |          171 |
|         2 | Accessories | Sunglasses     |          161 |
|         3 | Accessories | Belt           |          161 |
|         1 | Clothing    | Blouse         |          171 |
|         2 | Clothing    | Pants          |          171 |
|         3 | Clothing    | Shirt          |          169 |
|         1 | Footwear    | Sandals        |          160 |
|         2 | Footwear    | Shoes          |          150 |
|         3 | Footwear    | Sneakers       |          145 |
|         1 | Outerwear   | Jacket         |          163 |
|         2 | Outerwear   | Coat           |          161 |
+-----------+-------------+----------------+--------------+
11 rows in set (0.07 sec)
```

9. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

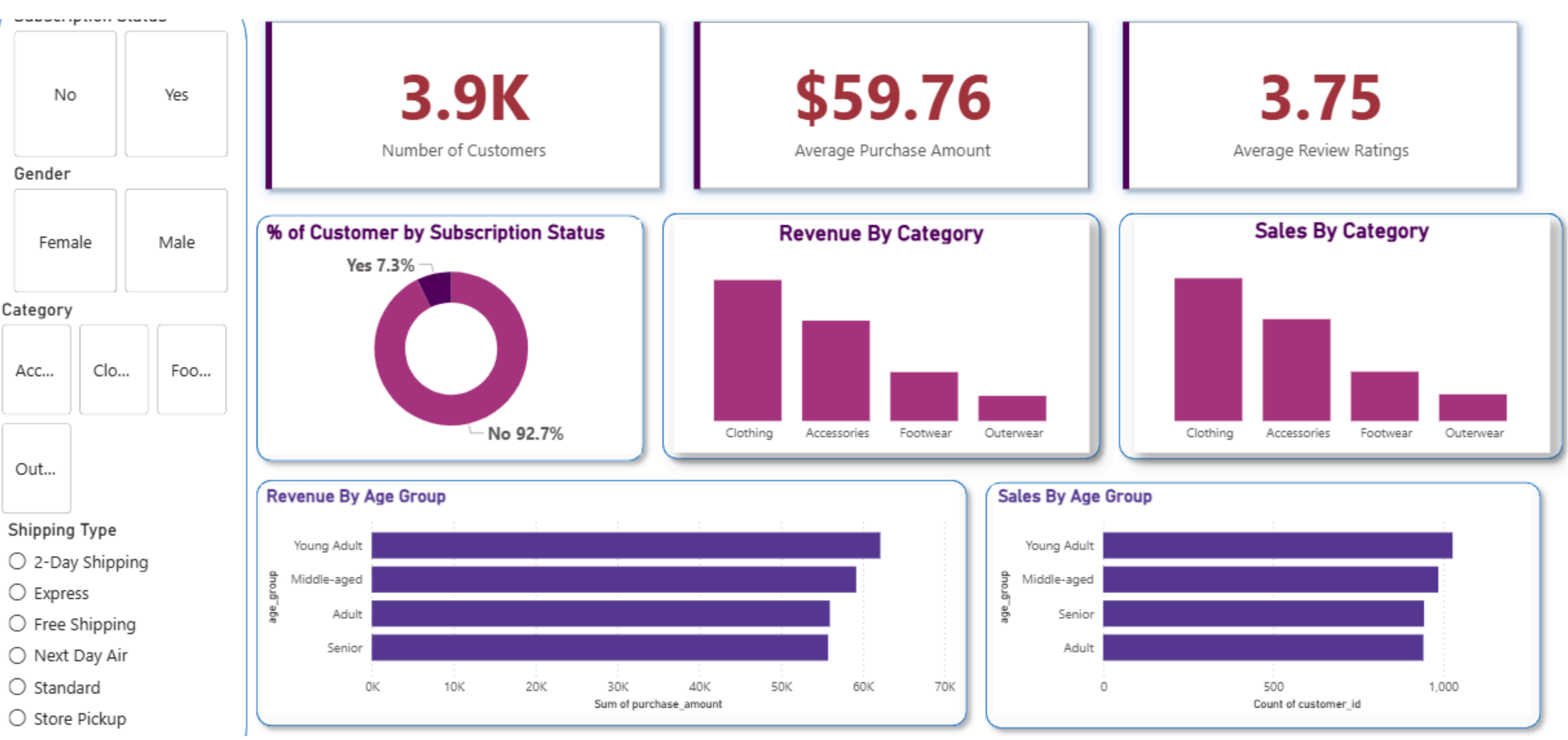
subscription_status	repeat_buyers
Yes	958
No	2518

10. **Revenue by Age Group** – Calculated total revenue contribution ofeach age group.

```
ysql> select age_group, SUM(purchase_amount) as total_revenue from customer group by age_group order by total_revenue desc;
+-------------+---------------+
| age_group   | total_revenue |
+-------------+---------------+
| Young Adult |         62143 |
| Middle-aged |         59197 |
| Adult       |         55978 |
| Senior      |         55763 |
+-------------+---------------+
rows in set (0.02 sec)
```

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.

- **Customer Loyalty Programs** – Reward repeat buyers to move them into the "Loyal" segment.
- **Review Discount Policy** – Balance sales boosts with margin control.
- **Product Positioning** – Highlight top-rated and best-selling products in campaigns.
- **Targeted Marketing** – Focus efforts on high-revenue age groups and express-shipping users.